

# Yi Hou (厚熠)

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Second-year Artificial Intelligence undergraduate working across the full LLM stack: Triton kernels, distributed training, pretraining data pipelines, and post-training. Author of five public technical reports built on controlled experiments.

## EDUCATION

**University of Chinese Academy of Sciences (UCAS)** — Beijing, China

B.Eng. in Artificial Intelligence · Sep 2025 – Jun 2029 (expected) · CET-6 (College English Test Band 6) passed

## SKILLS

- **Languages:** Python (primary), C/C++, JavaScript/TypeScript, LaTeX
- **LLM & Training:** PyTorch, Jittor; Transformer architectures from scratch; Triton kernels (FlashAttention-2); SFT / GRPO / DPO; RAG
- **Systems:** DDP / ZeRO-1 / FSDP implemented from scratch; mixed precision (bf16/fp16); Nsight Systems profiling
- **Engineering:** FastAPI, SQLite, Chroma, Git, Linux

## RESEARCH

### Prompt Dominance and Asymmetric Verifier Costs: Empirical Ablations of GRPO at 1B Scale on GSM8K

Single-author arXiv preprint (cs.LG, Oct 2026) · [arXiv:2610.04928](https://arxiv.org/abs/2610.04928)

- Controlled ablations at 1B scale (OLMo-2-0425-1B / GSM8K) establishing two findings: the prompt — not the advantage estimator — is the first-order variable (zero-shot 0.08% vs 3-shot 18.3%); and verifier noise is asymmetric — a 10%-flip verifier keeps 91–106% of RL’s attainable gain (two seeds) where best-of-32 selection keeps only 57%

## PROJECTS

### LLM from Scratch — 4 Projects · 5 Technical Reports

Public monorepo: [github.com/Helios-YQH/llm-from-scratch](https://github.com/Helios-YQH/llm-from-scratch)

- **Systems:** 5.7× faster attention than naive attention at 64K context (bf16) with hand-written Triton FlashAttention-2 kernels — fp32 64K contexts run where naive attention OOMs; from-scratch distributed stack: DDP variants (communication overlap cut the gradient-communication share from 69% to 59% on 4 GPUs), ZeRO-1 (~23.7% memory), FSDP (~43% steady-state memory, cross-checked against PyTorch’s official FSDP); Nsight profiling across 128M–3.4B-parameter models (bf16 mixed precision ~2–3×)
- **Pretraining:** trained a 22.7M-parameter decoder-only Transformer to 1.57 val loss in 1.5 GPU-hours on a 6×A6000 cluster at ICT CAS; bit-exact BPE tokenizer (4.18 bytes/token on TinyStories); controlled ablations over normalization / positional encoding / activation identified that a cosine LR tail can mask divergence
- **Data:** quantified crawled-corpus filtering with a paired ablation — -0.27 nats validation loss (perplexity 282 vs 367) — over a pipeline turning 2.14M Common Crawl documents into 340M GPT-2 tokens (HTML extraction, language ID, PII masking, quality classifiers, exact + MinHash-LSH dedup)
- **SFT + DPO:** quantified judge dependence in preference evaluation — AlpacaEval win rate for the same LoRA-SFT + DPO Llama-3.1-8B spans 3.8–9.4% under three frontier judges but 26–33% under a local 8B judge; measured MMLU / GSM8K capability trade-offs and SimpleSafetyTests safety

### ZhiyanNavigator — LLM-Powered Paper Learning Assistant

Backend developer · 2026 China International College Students’ Innovation Competition

- Built the backend learning loop of a paper-reading assistant: paper ingestion (arXiv link/ID or local PDF, with download and caching) → LLM-generated proficiency assessment → tiered study roadmap → card-by-card explanations with self-check quizzes
- Implemented RAG (local bge-small-zh embeddings + Chroma retrieval) and SSE-streaming LLM endpoints on FastAPI, which also serves the web UI; refactored an initially over-complex workflow into a minimal, reliable end-to-end pipeline

### Diffusion Models from Scratch — UNet vs. DiT, $\epsilon$ -Prediction vs. Rectified Flow (PyTorch)

Public repo: [github.com/Helios-YQH/diffusion-from-scratch](https://github.com/Helios-YQH/diffusion-from-scratch)

- Rectified flow beats  $\epsilon$ -prediction at every budget in a controlled UNet-vs-DiT comparison (best MNIST FID 0.88); DDIM at 50 steps matches the 1,000-step ancestral FID within 0.8% at ~20× the speed; negative results documented (DDIM degrades DiT); multi-GPU DDP over 5 GPUs, and a 174M-param CelebA UNet at FID 11.4

### Point Cloud Denoising — StraightPCF (CVPR 2024) Reproduction in Jittor

Solo · 6th Jittor AI Challenge, Apr–Jul 2026 — national top 100 of 3,668 · [github.com/Helios-YQH/jittor-competition-point-cloud](https://github.com/Helios-YQH/jittor-competition-point-cloud)

- Reproduced StraightPCF in Jittor — dynamic EdgeConv encoder, coupled velocity modules, and distance-modulated Euler integration (~530K parameters) predicting per-point displacements back onto the surface; trained on ~20K ShapeNet meshes, cutting Chamfer distance by 51% and point-to-surface error by 67% versus the noisy input (best competition score 67.44)

### Additional Projects

- **CUMCM 2026 (national mathematical-modeling contest, Problem A):** modeled 2D axisymmetric coupled heat-and-mass transfer for herb drying with implicit finite-difference solvers and nonlinear, state-dependent material properties; completed the modeling, numerical implementation, and LaTeX paper
- **3D Gaussian Splatting:** COLMAP reconstruction and 3DGS training pipeline